

TRANSPORT PHENOMENA II, CHE 312

EXAM 3

CLOSED BOOK, TIME: 120 MINUTES

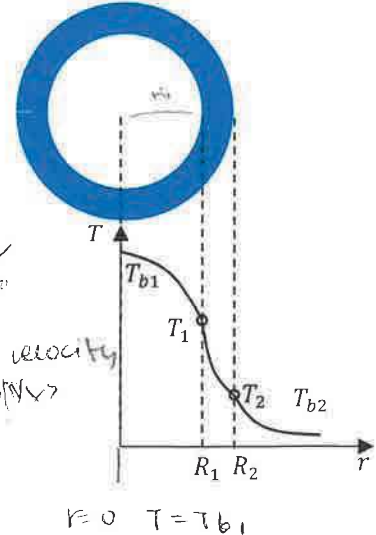
1. Consider a cylindrical pipe with an inner radius R_1 , an outer radius R_2 , and a constant thermal conductivity k . *no velocity*

- 1.1. Using the equation of energy in cylindrical coordinates, derive an equation for the steady-state temperature profile and the heat flux on the inner surface of the pipe in terms of T_1 and T_2 , the temperatures on the inner and outer surface of the pipe.

$$\rho \hat{C}_p \left(\cancel{\frac{\partial T}{\partial t}} + v \cancel{\frac{\partial T}{\partial r}} + \frac{v_\theta}{r} \cancel{\frac{\partial T}{\partial \theta}} + v \cancel{\frac{\partial T}{\partial z}} \right) = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v$$

$\downarrow$ no velocity
 $\mu \Phi_v \rightarrow$

- 1.2. A fluid with bulk temperature T_{b1} and heat transfer coefficient h_1 is in contact with the inner surface of the pipe while a second fluid with bulk temperature T_{b2} and heat transfer coefficient h_2 is in contact with the outer surface of the pipe. Derive an equation for the overall heat transfer coefficient U_1 for heat exchange between the two fluids with reference to the inner radius R_1 .



Hint: Heat flux on the inner surface is $q_r|_{r=R_1} = h_1(T_{b1} - T_1) = U_1(T_{b1} - T_{b2})$ and the one on the outer surface is $q_r|_{r=R_2} = h_2(T_2 - T_{b2}) = \frac{R_1}{R_2} q_r|_{r=R_1}$. $\Rightarrow h \Delta T = q$

$$q = -k \nabla T$$

$$q = -k \frac{dT}{dr}$$

$$1.1) \quad 0 = k \left[\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) \right]$$

$$0 = \frac{k}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) \Rightarrow \int \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0 \Rightarrow r \frac{dT}{dr} = C_1 \frac{dT}{dr} = \frac{C_1}{r} \frac{dT}{dr}$$

$$\Rightarrow \int \frac{dT}{dr} = \int \frac{C_1}{r} \Rightarrow T(r) = C_1 \log(r) + C_2$$

$$BC_1: \quad \frac{dT}{dr} = \frac{C_1}{r}$$

$$r=R_1 \quad T(R_1)=T_1 \Rightarrow T_1 = C_1 \log(R_1) + C_2$$

$$r=R_2 \quad T(R_2)=T_2 \Rightarrow T_2 = C_1 \log(R_2) + C_2$$

$$T_1 = \left(\frac{T_1 - T_2}{\log\left(\frac{R_1}{R_2}\right)} \right) \log R_1 + C_2$$

$$C_2 = \left(\frac{T_1 - T_2}{\log\left(\frac{R_1}{R_2}\right)} \right) \log R_2 - T_1$$

$$T_1 - T_2 = C_1 \log R_1 - C_1 \log R_2$$

$$T_1 - T_2 = C_1 \log\left(\frac{R_1}{R_2}\right)$$

$$C_1 = \frac{T_1 - T_2}{\log\left(\frac{R_1}{R_2}\right)}$$

Name: from 0 to $R_1 \Rightarrow$ Newton, $T_{b1} - T_1$

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R_2 to $T_{b2} \Rightarrow T_2 - T_{b2}$

from $R_1 \rightarrow R_2 \quad q = -k \nabla d$

$$\star (T_{b1} - T_1) = \frac{q_r|_{r=R_1}}{h_1} = \frac{q_r|_{r=R_1}}{U_1} \Rightarrow U_1 = \frac{(T_{b1} - T_1)}{q_r|_{r=R_1}}$$

$$\star (T_1 - T_2) = C_1 \log\left(\frac{R_1}{R_2}\right)$$

$$\star (T_2 - T_{b2}) = \frac{q_r|_{r=R_2}}{h_2} = \frac{R_1}{h R_2} \cdot q_r|_{r=R_1} = \frac{R_1}{h R_2} U_1 (T_{b1} - T_{b2})$$

$$= q_r|_{r=R_2} = \frac{(T_2 - T_{b2})}{\frac{h R_2}{R_1}}$$

Temp is constant.

$$(T_{b1} - T_{b2}) = \frac{q_r|_{r=R_1}}{U_1} + \frac{C_1 \log\left(\frac{R_1}{R_2}\right)}{U_1} + \frac{R_1}{h R_2} \cdot q_r|_{r=R_1}$$

$$2) \quad \frac{\partial C_A}{\partial t} = - \frac{d}{dy} \left(-D_{AB} \frac{dC_A}{dy} \right)$$

$$\frac{\partial C_A}{\partial t} = D_{AB} \frac{\partial^2 C_A}{\partial y^2} \Rightarrow \left(D_{AB} \frac{\partial^2 C_A}{\partial y^2} - \frac{\partial C_A}{\partial t} = 0 \right)$$

$$\theta = \frac{C_{A0} - C_A}{C_{A0}}$$

$$\eta = \frac{y}{b}$$

$$\Rightarrow C_{A0} - C_A = \theta C_{A0}$$

$$C_A = C_{A0} - \theta C_{A0}$$

$$\star C_A = C_{A0} (1 - \theta)$$

$$\star y = \eta b = (x dy)^2 = d\eta^2 b^2$$

$$\star t = \frac{\tau b^2}{D_{AB}}$$

$$dt = \frac{b^2}{D_{AB}} d\tau$$

$$i.c: t=0 \quad C_A=0, y=0 \quad \tau=0, \theta=1$$

$$BC \begin{cases} y=+b & C_A(+b) = C_{A0} \\ y=-b & C_A(-b) = C_{A0} \end{cases}$$

$$\eta=1 \quad \theta=0$$

$$\eta=-1 \quad \theta=0$$

$$\tau = \frac{D_{AB} t}{b^2}$$

[2.2]

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$$2) \quad D_{AB} \frac{\partial^2 C_A}{\partial y^2} - \frac{\partial C_A}{\partial t} = 0$$

$$D_{AB} \frac{\partial^2 (C_{A0}(1-\theta))}{\partial \eta^2 \cdot \frac{b^2}{D_{AB}}} - \frac{\partial (C_{A0}(1-\theta))}{\partial \tau \cdot \frac{b^2}{D_{AB}}} = 0$$

$$\left(\frac{D_{AB} C_{A0}}{b^2} \right) \left[\frac{\partial^2 (1-\theta)}{\partial \eta^2} - \frac{\partial (1-\theta)}{\partial \tau} \right] = 0 \Rightarrow \left(\frac{D_{AB} C_{A0}}{b^2} \right) \left[-\frac{\partial^2 \theta}{\partial \eta^2} + \frac{\partial \theta}{\partial \tau} \right] = 0$$

$$\Rightarrow \boxed{\frac{\partial^2 \theta}{\partial \eta^2} - \frac{\partial \theta}{\partial \tau} = 0} \quad \text{PDE (2.2)} \quad \Rightarrow \theta(\eta, \tau) = \sum_{n=0}^{\infty} f_n(\tau) \cos \left[\frac{2n+1}{2} \pi \eta \right]$$

$$c) \quad \theta(\eta, \tau) = \sum_0^{\infty} \underbrace{f_n(\tau)}_{\theta(\eta, \tau)} \cdot \underbrace{\cos \left[\frac{2n+1}{2} \cdot \pi \eta \right]}_{x(\eta)}$$

$$\theta(\eta, \tau) = f(\tau) \cdot x(\eta)$$

→ plug in PDE

$$\underbrace{\int \frac{1}{f} \frac{df}{d\tau}}_{f(\tau)} = \underbrace{\int \frac{1}{x} \frac{d^2 x}{d\eta^2}}_{x(\eta)}$$

$$f(\tau) =$$

$$x(\eta) =$$

$$\frac{\partial^2 (f(\tau) x(\eta))}{\partial \eta^2} - \frac{\partial (f(\tau) x(\eta))}{\partial \tau} = 0$$

$$x(\eta) \frac{\partial^2 f(\tau)}{\partial \tau} + f(\tau) \frac{\partial^2 x(\eta)}{\partial \eta^2} - \frac{\partial f(\tau) x(\eta)}{\partial \tau} - f(\tau) \frac{\partial x(\eta)}{\partial \tau} = 0$$

??? Don't know sorry :)

2.4) After I have my concentration profile $C_A(t, y)$

$$\text{flux on surface} : \int_{-b}^b N_{Ay} dy = \int_{-b}^b \left[-D_{AB} \frac{\partial}{\partial y} (C_A(t, y)) \right] dy$$

Name: